Tutorial_1_L13.pdf
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CSE 400: Fundamentals of Probability in Computing

Prepare a clear and exam-focused lecture for Lecture 13 (Tutorial). The document should be written so that a student can directly revise it for a closed-book or reading-based exam.

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The final document should be carefully organized, with revision notes that build intense conceptual clarity for exam preparation.

CSE 400: Fundamentals of Probability in Computing

Lecture 13 (Tutorial) – Exam-Oriented Revision Notes

These notes consolidate all definitions, distributions, assumptions, and standard probability tools used in Tutorial 1. The material is organized conceptually to strengthen revision for a closed-book or reading-based examination.

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1. Combinatorial Probability and Partitioning

Equal Partition of Distinct Objects

To divide 20 distinct dishes into four groups of five dishes each, where:

- Each group has exactly 5 dishes,
- The order within a group does not matter,
- The order of the groups does not matter,

we use multinomial counting with division by factorial terms to eliminate ordering.

Total number of ways:

$$\frac{20!}{(5!)^4 \cdot 4!}$$

- $(5!)^4$: removes ordering within each group.
- $4!$: removes ordering among groups.

Probability Under Structural Constraints

If there are five dishes from each of four cuisines, the favorable outcome corresponds to each platter containing dishes from exactly one cuisine.

Probability is computed as:

$$\frac{\text{Number of favorable partitions}}{\text{Total number of partitions}}$$

---

2. Continuous Uniform Distribution

Uniform Arrival Model

If a passenger arrives uniformly between 7:00 and 7:30 A.M., then:

$$X \sim \text{Uniform}(0, 30)$$

Probability over an interval:

$$P(a \leq X \leq b) = \frac{b-a}{30}$$

Since buses arrive every 15 minutes:

- Waiting time is determined by distance to next multiple of 15.
- Probabilities are computed via geometric intervals over total length 30.

---

3. Conditional Probability and Bayes' Theorem

Given:

$$P(Late) = 0.15$$

$$P(Early\ Leave) = 0.25$$

$$P(Late \cap Early\ Leave) = 0.08$$

We use:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Complement relationships:

$$P(A^c) = 1 - P(A)$$

$$P(A \cap B^c) = P(A) - P(A \cap B)$$

---

4. Poisson Distribution

Definition

A Poisson random variable with parameter λ:

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \quad k = 0, 1, 2, \ldots$$

Used when counting number of events in fixed interval with constant rate.

Expected Errors per Page

If expected number of typographical errors is 0.2:

$$\lambda = 0.2$$

(a) Probability of 0 errors:

$$P(X = 0) = e^{-0.2}$$

(b) Probability of 2 or more errors:

$$P(X \geq 2) = 1 - P(X = 0) - P(X = 1)$$

Monthly Airplane Crashes

Given:

$$\lambda = 3.5$$

(a) At least 2 accidents:

$$P(X \geq 2) = 1 - P(X = 0) - P(X = 1)$$

(b) At most 1 accident:

$$P(X \leq 1) = P(X = 0) + P(X = 1)$$

Conditional Poisson Probability

If $X \sim \text{Poisson}(3)$:

$$P(X \geq 3)$$

$$P(X \geq 3 | X \geq 1) = \frac{P(X \geq 3)}{P(X \geq 1)}$$

Using:

$$P(X \geq 1) = 1 - P(X = 0)$$

---

5. Normal Distribution and Q-Function

Given PDF:

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp\left(-\frac{(x+3)^2}{8}\right)$$

This corresponds to:

$$X \sim N(-3, 4)$$

Mean:

$$\mu = -3$$

Variance:

$$\sigma^2 = 4$$

Standardization:

$$Z = \frac{X - \mu}{\sigma}$$

Q-function:

$$Q(x) = P(Z > x)$$

All required probabilities must be expressed in terms of Q-functions after standardization.

Absolute value probabilities are converted into interval probabilities and then standardized.

---

6. Binomial Distribution and Jury Decision

Each juror independently makes correct decision with probability θ.

Let:

$$X \sim \text{Binomial}(12, \theta)$$

PMF:

$$P(X = k) = \binom{12}{k} \theta^k (1 - \theta)^{12-k}$$

Conviction occurs if:

$$X \geq 8$$

Probability of correct decision:

$$\sum_{k=8}^{12} \binom{12}{k} \theta^k (1 - \theta)^{12-k}$$

7. Poisson-Type PMF

Given:

$$p(i) = c\frac{\lambda^i}{i!}, \quad i = 0, 1, 2, \ldots$$

Since:

$$\sum_{i=0}^{\infty} p(i) = 1$$

We use:

$$\sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = e^{\lambda}$$

Thus:

$$c = e^{-\lambda}$$

Therefore:

$$P(X = 0) = e^{-\lambda}$$

$$P(X > 2) = 1 - \sum_{i=0}^{2} e^{-\lambda} \frac{\lambda^i}{i!}$$

---

8. Reliability and Majority Systems

Each component functions independently with probability p.

Let:

$$X \sim \text{Binomial}(n, p)$$

System operates if at least half function.

For:

- 5-component system → $X \geq 3$
- 3-component system → $X \geq 2$

Probability of operation:

$$P(X \geq \text{majority threshold})$$

General comparison:

$$(2k+1)\text{-component system vs } (2k-1)\text{-component system}$$

Compare:

$$P(X \geq k+1) \quad \text{and} \quad P(X \geq k)$$

using binomial sums.

Core Tools for Examination

The following mathematical structures are central:

1. Multinomial coefficients for partitioning
2. Uniform distribution over intervals
3. Conditional probability formula
4. Complement rule
5. Poisson PMF and cumulative complements
6. Normal standardization and Q-function
7. Binomial PMF and majority thresholds
8. Infinite series normalization using exponential function

These concepts form the foundational computational framework required to solve all tutorial problems.

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CSE 400: Fundamentals of Probability in Computing

Lecture 13 (Tutorial) – Consolidated Exam Revision Notes

These notes systematically organize all definitions, distributions, assumptions, identities, and standard probability results appearing in Tutorial-1 (Lecture 13). The presentation follows the mathematical structure underlying the problems and consolidates the tools required for solving them in an examination setting.

---

1. Combinatorial Structures and Multinomial Counting

Partitioning Distinct Objects

Given 20 distinct objects divided into four groups of five objects each, where:

- Each group contains exactly 5 objects,
- Order within each group is irrelevant,
- Order among the groups is irrelevant,

the number of possible divisions is

$$\frac{20!}{(5!)^4\,4!}.$$

- The factor $(5!)^4$ removes ordering within each group.
- The factor $4!$ removes ordering among the four groups.

Probability via Counting

If structural constraints are imposed (e.g., grouping by cuisine when exactly five objects of each type exist), probability is computed as

$$\frac{\text{Number of favorable partitions}}{\text{Total number of partitions}}.$$

---

2. Uniform Distribution on an Interval

A passenger arrives at a time uniformly distributed between 7:00 and 7:30 A.M.

Let $X \sim \text{Uniform}(0, 30)$.

For $a < b$,

$$P(a \le X \le b) = \frac{b-a}{30}.$$

When arrivals (e.g., buses every 15 minutes) determine waiting time, probabilities are evaluated by measuring the corresponding subinterval lengths relative to the total interval length 30.

---

3. Conditional Probability and Basic Identities

Let events A and B.

Conditional Probability

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0.$$

Complement Rule

$$P(A^c) = 1 - P(A).$$

Decomposition Identities

$$P(A \cap B^c) = P(A) - P(A \cap B).$$

These relations are applied when probabilities of joint and marginal events are given.

---

4. Poisson Distribution

Definition

A random variable X is Poisson with parameter $\lambda > 0$ if

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \quad k = 0, 1, 2, \ldots$$

Key Properties

- Mean: $E[X] = \lambda$.
- Used for modeling counts over fixed intervals.

Cumulative and Complement Forms

$$P(X \geq m) = 1 - \sum_{k=0}^{m-1} \frac{e^{-\lambda}\lambda^k}{k!}$$

$$P(X \leq m) = \sum_{k=0}^{m} \frac{e^{-\lambda}\lambda^k}{k!}$$

Conditional Probability for Poisson

$$P(X \geq a \mid X \geq b) = \frac{P(X \geq a)}{P(X \geq b)}, \quad a \geq b.$$

---

5. Exponential Series Identity

The identity

$$\sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = e^{\lambda}$$

is used for normalization of probability mass functions of the form

$$p(i) = c\frac{\lambda^i}{i!}, \quad i = 0, 1, 2, \ldots$$

Imposing

$$\sum_{i=0}^{\infty} p(i) = 1$$

gives

$$c = e^{-\lambda}.$$

---

6. Binomial Distribution and Majority Rules

Definition

If each of n independent trials succeeds with probability p, then

$$X \sim \text{Binomial}(n, p)$$

with

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

Majority Threshold

For systems requiring at least half the components to function:

- If $n = 5$, system operates if $X \geq 3$.
- If $n = 3$, system operates if $X \geq 2$.
- In general, if $n = 2k + 1$, operation requires $X \geq k + 1$.

Probability of operation:

$$P(X \geq \text{threshold}) = \sum_{k=\text{threshold}}^{n} \binom{n}{k} p^k (1-p)^{n-k}.$$

System comparisons are made by comparing these cumulative binomial probabilities.

---

7. Jury Decision Model

Assume:

- 12 independent jurors,
- Each makes the correct decision with probability θ.

Let

$$X \sim \text{Binomial}(12, \theta).$$

A conviction requires at least 8 guilty votes:

$$P(\text{correct decision}) = \sum_{k=8}^{12} \binom{12}{k} \theta^k (1-\theta)^{12-k}.$$

Independence and identical probability assumptions are essential.

---

8. Normal Distribution and Q-Function

Given density

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp\left(-\frac{(x+3)^2}{8}\right),$$

this corresponds to a normal distribution with:

$$\mu = -3, \quad \sigma^2 = 4.$$

Standardization

$$Z = \frac{X-\mu}{\sigma}.$$

Q-Function

$$Q(x) = P(Z > x).$$

All required probabilities are expressed in terms of the Q-function after standardization.

Absolute value expressions are converted into interval probabilities and then standardized accordingly.

Core Mathematical Tools for Examination

The tutorial relies on the following foundational structures:

- Multinomial counting for equal partitions

- Uniform distribution on finite intervals
- Conditional probability and complement identities
- Poisson distribution and cumulative complements
- Exponential series normalization
- Binomial distribution with majority thresholds
- Normal distribution standardization and Q-function

These tools form the complete conceptual and computational framework required for solving all problems in Lecture 13 (Tutorial).

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CSE 400: Fundamentals of Probability in Computing

Lecture 13 (Tutorial) – Structured Exam Revision Notes

This document consolidates all mathematical definitions, distributions, assumptions, identities, and standard results appearing in Tutorial-1 (Lecture 13). The material is organized conceptually and follows strictly the content reflected in the tutorial sheet. No additional results or external explanations are introduced.

---

1. Combinatorial Probability and Equal Partitions

Multinomial Counting for Equal Groups

Given 20 distinct objects divided into four groups of five objects each, with:

- exactly five objects per group,
- order within each group irrelevant,
- order of the groups irrelevant,

the number of possible divisions is:

$$\frac{20!}{(5!)^4\,4!}.$$

- The term $(5!)^4$ eliminates permutations within each group.
- The term $4!$ eliminates permutations among the groups.

Probability via Counting Principle

When structural restrictions are imposed (e.g., five objects from each category), probability is computed as:

$$P(\text{event}) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}.$$

---

2. Continuous Uniform Distribution

If arrival time is uniformly distributed over an interval $[0, 30]$:

$$X \sim \text{Uniform}(0, 30),$$

then for any $a < b$,

$$P(a \leq X \leq b) = \frac{b-a}{30}.$$

When periodic arrivals occur at fixed intervals (e.g., every 15 minutes), waiting-time probabilities are obtained by identifying appropriate subinterval lengths relative to total interval length 30.

---

3. Conditional Probability Framework

For events A and B, with $P(B) > 0$,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Complement Rule

$$P(A^c) = 1 - P(A).$$

Decomposition Identity

$$P(A \cap B^c) = P(A) - P(A \cap B).$$

These relations are used when joint and marginal probabilities are specified.

---

4. Poisson Distribution

Definition

A Poisson random variable with parameter $\lambda > 0$ has probability mass function:

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \quad k = 0, 1, 2, \ldots$$

Cumulative Forms

$$P(X \le m) = \sum_{k=0}^{m} \frac{e^{-\lambda}\lambda^k}{k!},$$

$$P(X \ge m) = 1 - \sum_{k=0}^{m-1} \frac{e^{-\lambda}\lambda^k}{k!}.$$

Conditional Probability for Poisson

$$P(X \ge a \mid X \ge b) = \frac{P(X \ge a)}{P(X \ge b)}, \quad a \ge b.$$

---

5. Exponential Series Identity and Normalization

The series identity:

$$\sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = e^{\lambda}$$

is used to determine the normalizing constant c in probability mass functions of the form:

$$p(i) = c\frac{\lambda^i}{i!}, \quad i = 0, 1, 2, \ldots$$

Imposing:

$$\sum_{i=0}^{\infty} p(i) = 1$$

gives:

$$c = e^{-\lambda}.$$

---

6. Binomial Distribution

Definition

If each of n independent trials results in success with probability p, then:

$$X \sim \text{Binomial}(n, p)$$

with probability mass function:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \ldots, n.$$

Majority Threshold

If system operation requires at least half of components functioning:

- For $n = 5$: operation if $X \geq 3$.
- For $n = 3$: operation if $X \geq 2$.
- For $n = 2k + 1$: operation if $X \geq k + 1$.

Probability of effective operation:

$$P(X \geq \text{threshold}) = \sum_{k=\text{threshold}}^{n} \binom{n}{k} p^k (1-p)^{n-k}.$$

Comparisons between systems involve comparing these cumulative binomial sums.

---

7. Jury Decision Model

Assumptions:

- 12 independent jurors,
- Each juror makes the correct decision with probability θ.

Let:

$$X \sim \text{Binomial}(12, \theta).$$

Conviction requires at least 8 guilty votes:

$$P(\text{correct decision}) = \sum_{k=8}^{12} \binom{12}{k} \theta^k (1-\theta)^{12-k}.$$

Independence and identical probability are assumed.

---

8. Normal Distribution and Q-Function

Given density:

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp\left(-\frac{(x+3)^2}{8}\right),$$

this corresponds to a normal distribution with:

$$\mu = -3, \quad \sigma^2 = 4.$$

Standardization

$$Z = \frac{X - \mu}{\sigma}.$$

Q-Function Definition

$$Q(x) = P(Z > x).$$

All probabilities are expressed in terms of Q-functions after standardization.

Absolute value expressions are converted into equivalent interval statements before applying standardization.

Consolidated Mathematical Framework

Lecture 13 relies exclusively on the following core tools:

- Multinomial counting for equal partitions
- Probability via counting principle
- Continuous uniform distribution on finite interval
- Conditional probability and complement identities
- Poisson distribution and cumulative complements
- Exponential series identity for normalization
- Binomial distribution and majority thresholds
- Normal distribution standardization and Q-function representation

These results form the complete mathematical foundation necessary for solving all tutorial problems under examination conditions.

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CSE 400: Fundamentals of Probability in Computing

Lecture 13 (Tutorial) – Comprehensive Exam Revision Notes

The following notes consolidate all definitions, assumptions, probability models, identities, and standard results that arise in Tutorial-1 (Lecture 13). The material is organized conceptually and strictly reflects the content of the tutorial sheet. No additional theory or examples are introduced. The presentation is structured to support direct revision for a closed-book examination.

1. Counting Methods and Equal Partitions

Division of Distinct Objects into Equal Groups

When 20 distinct objects are divided into four groups of five objects each, under the conditions that:

- each group contains exactly five objects,
- order within each group is irrelevant,
- order among the groups is irrelevant,

the total number of possible divisions is

$$\frac{20!}{(5!)^4\,4!}.$$

- The factor $(5!)^4$ removes permutations within each group.
- The factor $4!$ removes permutations among the four groups.

Probability via Counting

For any structural restriction on partitions,

$$P(\text{event}) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}.$$

---

2. Uniform Distribution on a Finite Interval

If a random arrival time is uniformly distributed between 7:00 and 7:30, let

$$X \sim \text{Uniform}(0, 30).$$

For $a < b$,

$$P(a \le X \le b) = \frac{b-a}{30}.$$

When events (such as arrivals every 15 minutes) determine waiting times, probabilities are computed by identifying the corresponding subinterval lengths relative to the total interval length 30.

---

3. Conditional Probability and Fundamental Identities

For events A and B, with $P(B) > 0$,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Complement Rule

$$P(A^c) = 1 - P(A).$$

Intersection Decomposition

$$P(A \cap B^c) = P(A) - P(A \cap B).$$

These relations are applied when marginal and joint probabilities are provided.

---

4. Poisson Distribution

Definition

A random variable X is Poisson with parameter $\lambda > 0$ if

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \quad k = 0, 1, 2, \ldots$$

Cumulative Probabilities

$$P(X \leq m) = \sum_{k=0}^{m} \frac{e^{-\lambda}\lambda^k}{k!},$$

$$P(X \geq m) = 1 - \sum_{k=0}^{m-1} \frac{e^{-\lambda}\lambda^k}{k!}.$$

Conditional Probability

$$P(X \geq a \mid X \geq b) = \frac{P(X \geq a)}{P(X \geq b)}, \quad a \geq b.$$

---

5. Exponential Series and Normalization of PMF

The identity

$$\sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = e^{\lambda}$$

is used to determine the constant c in

$$p(i) = c\frac{\lambda^i}{i!}, \quad i = 0, 1, 2, \ldots$$

Imposing

$$\sum_{i=0}^{\infty} p(i) = 1$$

gives

$$c = e^{-\lambda}.$$

---

6. Binomial Distribution

Definition

If each of n independent trials succeeds with probability p,

$$X \sim \text{Binomial}(n, p),$$

with probability mass function

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \ldots, n.$$

Majority Requirement

If a system operates when at least half the components function:

- For $n = 5$: operation if $X \geq 3$.
- For $n = 3$: operation if $X \geq 2$.
- For $n = 2k + 1$: operation if $X \geq k + 1$.

Probability of effective operation:

$$P(X \geq \text{threshold}) = \sum_{k=\text{threshold}}^{n} \binom{n}{k} p^k (1-p)^{n-k}.$$

System comparisons are made by comparing these cumulative binomial probabilities.

---

7. Jury Decision Model

Assumptions:

- 12 jurors,

- independence among jurors,
- each juror makes the correct decision with probability θ.

Let

$$X \sim \text{Binomial}(12, \theta).$$

Conviction requires at least 8 guilty votes:

$$P(\text{correct decision}) = \sum_{k=8}^{12} \binom{12}{k} \theta^k (1-\theta)^{12-k}.$$

---

8. Normal Distribution and Q-Function

Given the probability density function

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp\left(-\frac{(x+3)^2}{8}\right),$$

this corresponds to a normal distribution with

$$\mu = -3, \qquad \sigma^2 = 4.$$

Standardization

$$Z = \frac{X-\mu}{\sigma}.$$

Q-Function

$$Q(x) = P(Z > x).$$

All required probabilities are expressed in terms of Q-functions after standardization.

Absolute value expressions are first rewritten as interval probabilities and then standardized.

Integrated Mathematical Framework for Lecture 13

The tutorial integrates the following core mathematical structures:

- Multinomial counting for equal partitions
- Probability as ratio of favorable to total outcomes
- Uniform distribution on finite intervals

- Conditional probability and complement identities
- Poisson distribution and cumulative complements
- Exponential series normalization
- Binomial distribution and majority thresholds
- Normal distribution and Q-function representation

These results collectively constitute the complete theoretical and computational foundation required for solving all problems in Lecture 13 under examination conditions.